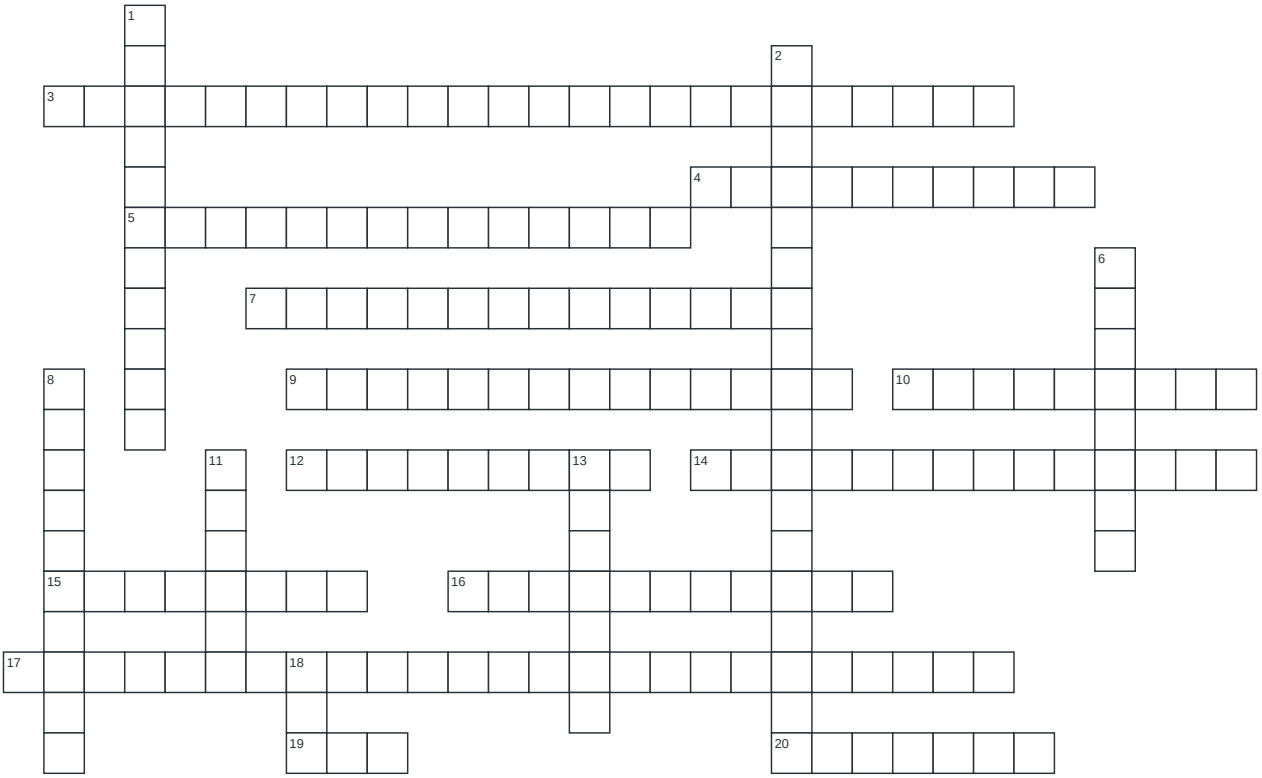
Cell Structure and Function Crossword

Hard | 20 clues | Student Worksheet

Name: _____ Date: _____

Directions: Use the clues to complete the grid. Omit spaces, punctuation, and hyphens.



ACROSS

3. Relationship between a cell's exchange surface and its internal volume.

4. Living organisms consist of cells, and cells arise from existing cells

5. Plant-cell compartment that stores water and helps maintain pressure

7. Boundary that regulates exchange between a cell and its surroundings

9. Process that uses light energy to build sugars from carbon dioxide and water.

10. Cell contents outside the nucleus and inside the plasma membrane

12. Net movement of particles from higher to lower concentration.

14. Organelle that modifies, sorts, and packages many proteins

15. Cell structure that assembles proteins

16. Organelle where photosynthesis occurs in plants and algae

ACROSS (continued)

17. Membrane network with attached ribosomes that helps process proteins

19. Molecule that is a major short-term energy-transfer carrier in cells

20. Membrane-bound organelle that contains most eukaryotic DNA

DOWN

1. Measure of the total outer area available for exchange.

2. Process by which cells develop structures suited to particular functions.

6. Rigid outer layer that supports many plant, fungal, and bacterial cells

8. A prokaryotic cell lacks a membrane-bound nucleus

11. Group of similar cells and associated material performing related functions

13. Net movement of water across a selectively permeable membrane.

18. Molecule that stores hereditary information in most cells

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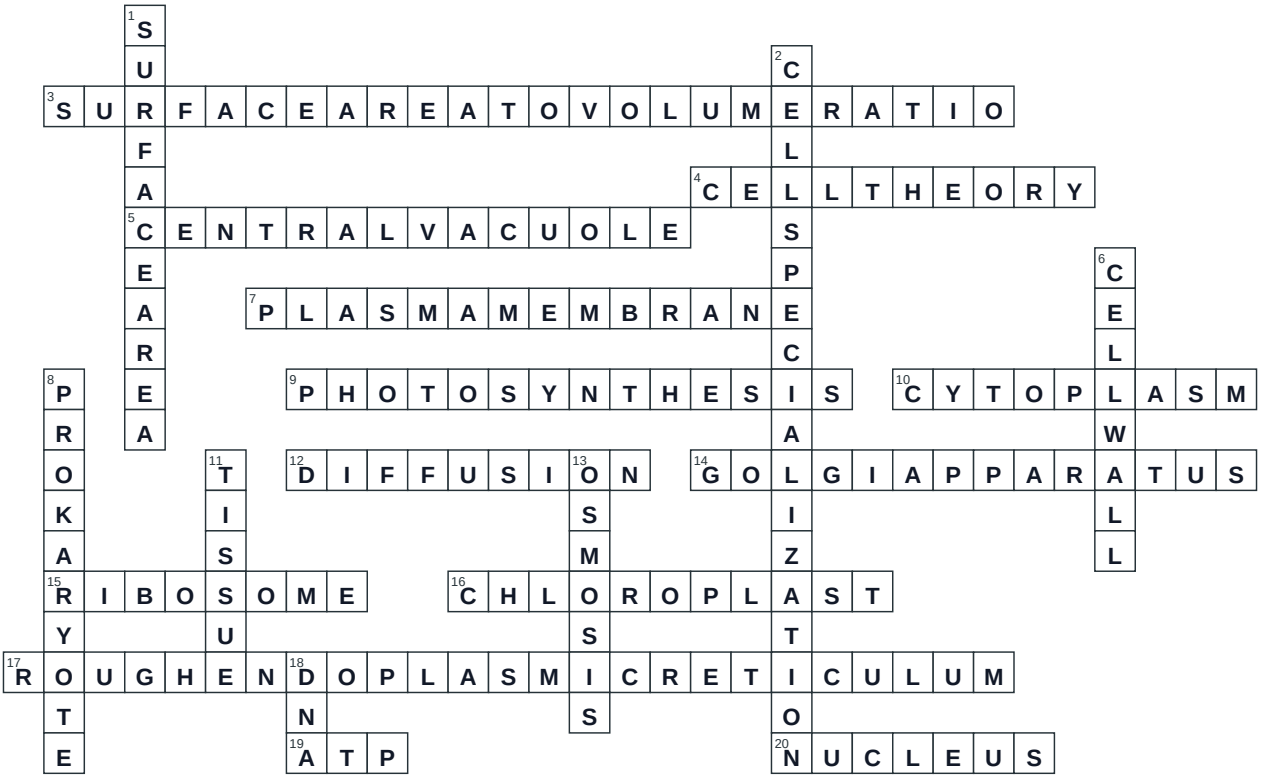
Hard | 20 clues | Teacher Answer Key

TEACHER ANSWER KEY

Completed grid and numbered answers. Spaces and punctuation are omitted in the grid.



ACROSS

3. Surface Area-to-Volume Ratio
4. Cell Theory
5. Central Vacuole
7. Plasma Membrane
9. Photosynthesis
10. Cytoplasm
12. Diffusion
14. Golgi Apparatus
15. Ribosome
16. Chloroplast
17. Rough Endoplasmic Reticulum

ACROSS (continued)

19. ATP
20. Nucleus

DOWN

1. Surface Area
2. Cell Specialization
6. Cell Wall
8. Prokaryote
11. Tissue
13. Osmosis
18. DNA

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